

Caner Derici

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Technical Skills

Areas of Experience:	Compilers & Language Runtimes · Machine Learning · Distributed Systems
Languages:	C++ · LLVM · MLIR · Go · Python · Racket · x86_64 ASM · MIR/TableGen · CUDA C++ · PTX
Workflows & Productivity:	Linux · Neovim · tmux · VSCode · OpenCode · Codex · Git · GH Actions · Obsidian
APIs, DB & Misc:	REST · gRPC · GraphQL · OpenAPI · FastAPI · DQLite · MongoDB · PostgreSQL · CI/CD

Education

Ph.D., Indiana University, Bloomington , Computer Science, Compilers	2015 – 2025
Dissertation: Self-Hosting Functional Programming Languages on Meta-Tracing JIT Compilers	
M.Sc., Boğaziçi University , Computer Engineering, Machine Learning	2012 – 2014
B.Sc., Bilgi University , Computer Science, Compilers	2005 – 2010

Selected Projects

Pycket: A tracing JIT compiler for full-scale bootstrapping functional language

Built the first tracing JIT capable of running a self-hosting language on a meta-tracing JIT backend, and worked on it for over five years during my PhD. Created a new IR and implemented performance analysis tools, run-time optimizations, and formalisms to improve throughput. Added code-gen for the FFI layer, engines, and meta-continuations for user threads. Used Python, C++, and Racket.

Around the World in Compilers

Designed and implemented a growing collection of experimental compilers, one for each letter of the alphabet. Built shared frontend pipelines targeting LLVM IR, MLIR, NVPTX, and LLVM MIR. Explored MLIR affine/linalg lowering, CUDA/NVPTX kernel code-gen, MLIR.smt + Z3 runtime verification, and out-of-tree LLVM MachineScheduler plugins for MIR scheduling experiments, as well as LLVM ORC JIT compilation. In C++23, LLVM/MLIR 20.

Racket IR: Linklets

Co-designed the IR that ships in the main Racket distribution (see publications). Documented its design, and developed its formal semantics in my PhD dissertation.

Abstract Machine Interp: An executable theoretical model

Executable theoretical computation model that investigates switching between heap allocated continuations and native stack frames to reduce memory pressure in CESK machines. In PLT Redex.

Rax: A full-stack nanopass compiler from Racket to x86_64 ISA

Implemented all the passes (closure conversion, register allocation, code-gen, etc.), along with garbage collection. Developed optimizations, such as inlining, loop-invariant code motion, and proper tail calls. In Racket and C++14.

FARS: Functional Automated Reasoning System

A resolution/refutation theorem prover, for expressions in first-order predicate logic with equality. Used binary paramodulation, and forward and backward subsumption for equational deduction. In Racket.

Hazırcevap (Witty): A closed-domain question-answering system for high school students

Government-funded large-scale question-answering system. M.Sc. thesis on NLP. Led the R&D team (3 faculty members, 4 grad students). Developed a Hidden Markov Random Field model for question analysis, and relevance metrics for information retrieval and response generation (see publications). In Python and JavaScript.

Experience

Canonical USA	REMOTE, US
SWE II (L5 - IC4) · DSL Compiler · Distributed Orchestration at Scale	2021 – 2024
- Designed and developed a custom scripting DSL for (CPU & memory) restricted computations. Built compiler on top of a modified Lua interpreter in Go. - Developed and maintained Juju, an eventually consistent distributed orchestration system used by ~200 companies globally, capable of handling 1000+-node workloads with 99.9% availability on any infrastructure (Kubernetes or otherwise) across various cloud providers (e.g., AWS, GCE). All in Go. - Improved reliability and fault tolerance by implementing machine provision services on relational DQLite state back-end, migrating from NoSQL MongoDB (e.g., sample PR). - Owned client libraries for three years—python-libjuju, Terraform Juju Provider; doubled active users and maintained a steady release cadence. - Took part in roadmap planning, coordinated cross-team work; mentored junior engineers, and created structured bootcamp process to improve hiring and onboarding.	

Indiana University

IN, US

PhD Work · Research Assistant · Course Instructor

2015 – 2021

- Independently took an ambiguous, uncharted compiler problem to a working product; built the first-ever tracing JIT compiler that is a full-scale runtime for a self-hosting, production-grade language. Conducted a full performance investigation and designed new optimization algorithms (see [PhD dissertation](#)).
- Taught data structures & algorithms, compilers, virtual machines, and domain specific languages.

Asseco SEE Group

Software Engineer

2010-2012

- International software company developing virtual payment systems for e-commerce platforms. I developed and delivered 3 virtual point-of-sale projects in 2 years. Used Java, Apache Tomcat, Spring, Mercurial, Jira.

Selected Publications

- Flatt M., Derici C. Dybvig R. K., Keep A. et. al. “Rebuilding racket on chez scheme”, ICFP’19
- Derici C. et. al. “A closed-domain question answering framework using reliable resources to assist students” Natural Language Engineering Journal’18
- Derici C. et. al. “Question analysis for a closed domain question answering system”, CICLING’15
- Derici C. et. al. “Rule-based focus extraction in question answering systems”, IEEE SIU’14
- Başar R. E., Derici C., and Şenol Ç. “World With Web: A compiler from world applications to JavaScript”. Scheme and Functional Programming Workshop’09

Awards & Scholarships

- Scholarship and award for a project on teaching natural languages to hearing impaired, 2014.
 - Full Scholarship for PhD, 2015-2025
 - Full Scholarship for MSc, 2012
 - Full Scholarship for BSc, 2005-2010
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